MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 4.7 Notes

Coordinates and Change of Basis

Devotional Thought

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Coordinate Representation in R^n

In section 4.5 we saw that if B is a basis for a vector space V, then every vector $\mathbf{x}$ in V can be expressed in one and only one way as a linear combination of vectors in B:

Given a basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ for a vector space V, for every vector $\mathbf{x}$ in V there exist *unique* coefficients c_1, c_2, ..., c_n such that $\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$.

The unique coefficients in the linear combination are called the **coordinates of x relative to B**. In the context of coordinates, the order of the vectors in the basis is important, so this will sometimes be emphasized by referring to the basis B as an *ordered* basis.

Definition Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be an ordered basis for a vector space V and let $\mathbf{x}$ be a vector in V such that

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n.$$

The scalars c_1, c_2, ..., c_n are the **coordinates of x relative to the basis B**. The **coordinate matrix** (or **coordinate vector**) **of x relative to B** is the column matrix in R^n whose components are the coordinates of $\mathbf{x}$.

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Remark Given a vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ in R^n, the components x_i are the coordinates of $\mathbf{x}$ *relative to the standard basis S for R^n*.

Example [Example 1]
Find the coordinate matrix of $\mathbf{x} = (-2, 1, 3)$ in R^3 relative to the standard basis

$$S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

Example [Example 2] The coordinate matrix of $\mathbf{x}$ in R^2 relative to the (nonstandard) ordered basis $B = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(1, 0), (1, 2)\}$ is

$$[\mathbf{x}]_B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Find the coordinate matrix of $\mathbf{x}$ relative to the standard basis $B' = \{\mathbf{u}_1, \mathbf{u}_2\} = \{(1, 0), (0, 1)\}$.

Example [Example 3] Find the coordinate matrix of $\mathbf{x} = (1, 2, -1)$ in R^3 relative to the (nonstandard) basis

$$B' = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \{(1, 0, 1), (0, -1, 2), (2, 3, -5)\}.$$

Remark It would be incorrect in the above example to write the coordinate matrix as

$$\mathbf{x} = \begin{bmatrix} 5 \\ -8 \\ -2 \end{bmatrix}.$$

Why?

The procedure demonstrated in Examples 2 and 3 is called a **change of basis**. That is, we were given the coordinates of a vector relative to a basis B and were asked to find the coordinates relative to another basis B'.

For instance, if in Example 3 we let B be the standard basis, then the problem of finding the coordinate matrix of $\mathbf{x} = (1, 2, -1)$ relative to the basis B' becomes one of solving for c_1, c_2, and c_3 in the matrix equation

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 3 \\ 1 & 2 & -5 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}.$$

The square matrix is the *transition matrix from B' to B*, usually denoted P where the column matrix on the left side of the equation is $[\mathbf{x}]_{B'}$, coordinate matrix of $\mathbf{x}$ with respect to B', while the column matrix on the right is $[\mathbf{x}]_B$, the coordinate matrix relative to B. Multiplication by the transition matrix P changes a coordinate matrix relative to B' into a coordinate matrix relative to B. That is,

$$P\,[\mathbf{x}]_{B'} = [\mathbf{x}]_B\,.$$

To go backwards and perform a change of basis from B to B', we use the matrix P^{-1}, which is the the *transition matrix from B to B'*:

$$[\mathbf{x}]_{B'} = P^{-1}\,[\mathbf{x}]_B\,.$$

So, the change in basis problem in Example 3 can be represented by the matrix equation

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -1 & 4 & 2 \\ 3 & -7 & -3 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -8 \\ -2 \end{bmatrix}.$$

Change of Basis in R^n

Generalizing the preceding discussion, assume that

$$B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} \quad \text{and} \quad B' = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$$

are two ordered bases for R^n. If $\mathbf{x}$ is a vector in R^n and

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \quad \text{and} \quad [\mathbf{x}]_{B'} = \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{bmatrix}$$

are the coordinate matrices of $\mathbf{x}$ relative to B and B', then the **transition matrix P from B' to B** is the matrix P such that

$$[\mathbf{x}]_B = P\,[\mathbf{x}]_{B'}\,.$$

The next theorem also tells us that the **transition matrix from B to B'** must be P^{-1}. That is,

$$[\mathbf{x}]_{B'} = P^{-1}\,[\mathbf{x}]_B\,.$$

Theorem 4.20 The Inverse of a Transition Matrix

If P is the transition matrix from a basis B' to a basis B in R^n, then P is invertible and the transition matrix from B to B' is P^{-1}.

The book contains a proof of this fact.

Theorem 4.21 **Transition Matrix from B to B'**

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $B' = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be two bases for R^n. Then the transition matrix P^{-1} from B to B' can be found by using Gauss-Jordan elimination on the $n \times 2n$ augmented matrix $[B' \mid B]$ to yield $\left[I_n \mid P^{-1}\right]$ where I_n is the $n \times n$ identity matrix.

Example [Example 4] Find the transition matrix from B to B' for the bases for R^3 below.

$$B = \{(1,0,0),(0,1,0),(0,0,1)\} \quad \text{and} \quad B' = \{(1,0,1),(0,-1,2),(2,3,-5)\}$$

Class Discussion

1. Let $B = \{(1,0),(1,2)\}$ and $B' = \{(1,0),(0,1)\}$. Form the matrix $[B' \mid B]$.

2. Make a conjecture about the necessity of using Gauss-Jordan elimination to obtain the transition matrix P^{-1} when the change of basis is from a nonstandard basis to a standard basis.

Example [Example 5] Find the transition matrix from B to B' for the bases for R^2 below.

$$B = \{(-3,2),(4,-2)\} \quad \text{and} \quad B' = \{(-1,2),(2,-2)\}$$

Find the transition matrix from B' to B.

Remark The transition matrix from B' to B is actually $P = B^{-1}B'$. The transition matrix from B to B' is thus $P^{-1} = (B')^{-1}B$. These relationships can be used to check the results obtained from Gauss-Jordan elimination.

Example [Exercise 39]

Consider the given bases B and B', as well as the coordinate matrix $[\mathbf{x}]_{B'}$

$$B = \{(1, 0, 2), (0, 1, 3), (1, 1, 1)\}$$

$$B' = \{(2, 1, 1), (1, 0, 0), (0, 2, 1)\}$$

$$[\mathbf{x}]_{B'} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

(a) Find the transition matrix from B to B'.

(b) Find the transition matrix from B' to B.

(c) Verify that the two transition matrices are inverses of each other.

(d) Find the coordinate matrix $[\mathbf{x}]_B$, given the coordinate matrix $[\mathbf{x}]_{B'}$.